mimic

Imitates Machines, Inspects Code

User Guide

Running the emulator: desktop, command line, media, controllers, live preview and debugger

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1 Introduction

mimic is a cycle-accurate emulator for classic home computers and game consoles, built on one shared core. Every machine gets the same remote debugger and the same live preview stream automatically: you can watch a machine in the browser, halt it at a breakpoint, inspect its chips and step through its code, whichever machine it is.

This guide describes how to run the emulator: building it, supplying ROMs, starting machines in the desktop window or on the command line, inserting disks, tapes and cartridges, using keyboards, joysticks, gamepads and mice, choosing a monitor look, and connecting the browser preview and debugging tools.

1.1 Supported machines

Machine id	Machine	Models (--model)
`c64`	Commodore 64	`pal` (default), `ntsc`
`vic20`	Commodore VIC-20	`pal`, `ntsc`
`pet`	Commodore PET	`4032` (default), `2001`, `3032`, `8032`
`c16`, `plus4`	Commodore 16, Plus/4	`pal`, `ntsc`
`c128`	Commodore 128 (40 and 80 columns, Z80)	`pal`, `ntsc`
`atari8`	Atari 400/800/800XL/65XE/130XE	`800xl` (default), `65xe`, `130xe`, `800`, `400`; each also `-ntsc`
`atari2600`	Atari 2600 (VCS)	`auto` (default: NTSC or PAL from the program's field length), `ntsc`, `pal`
`cpc`	Amstrad CPC	`6128` (default), `664`, `464`
`amiga`	Commodore Amiga 500/1000/2000 (OCS), 500+ (ECS), 1200 (AGA)	`500` (default), `1000`, `2000`, `500plus`, `1200`; each also `-ntsc`
`zx81`	Sinclair ZX81 / Timex Sinclair 1000	`zx81` (default, PAL), `ts1000` (NTSC)
`spectrum`	Sinclair ZX Spectrum	`48k` (default), `128k`, `plus2`
`coleco`	ColecoVision	`ntsc` (default), `pal`
`nes`	Nintendo Entertainment System / Famicom	`ntsc` (default), `pal`
`snes`	Super Nintendo / Super Famicom	`ntsc` (default), `pal`
`vectrex`	Vectrex (vector display)	`vectrex`
`apple2`	Apple II, II+, IIe	`iie` (enhanced IIe, default), `iie-unenhanced`, `iiplus`, `ii`
`mac`	Apple Macintosh 128K, 512K, Plus, SE	`plus` (default), `128k`, `512k`, `se`
`sim6502`	6502 reference machine for tests	`default`

`mimic-headless --list` prints the same list from the program itself.

1.2 The two programs

Program	Purpose
`mimic-desktop`	The emulator with a window, sound, keyboard, gamepads and mouse. It can also serve the debugger, the VICE monitor and the browser preview.
`mimic-headless`	No window and no sound: run a number of frames and save a picture, record sound to a WAV file, write savestates, or serve the debugger and the preview. For scripts, tests and remote use.

2 Installation and quick start

2.1 Requirements

- **Rust** (stable toolchain) with `cargo` — install it from rustup.rs.
- **Linux only:** the packages `libasound2-dev` and `libudev-dev` for the desktop program (sound and gamepads).
- **Optional:** Python 3 for the debug client `tools/polydbg.py` (standard library only).

2.2 Building

Always build the optimised (release) version; a debug build is far slower than real time.

```
cargo build --release -p emu-desktop     # the desktop emulator
cargo build --release -p emu-headless    # the command-line frontend
```

The programs end up in `target/release/` — on Windows `mimic-desktop.exe` and `mimic-headless.exe`. They need no other files and can be copied anywhere; ROMs are found through `--rom-dir` or the environment variable `EMU_ROM_DIR`.

> **Watch out.** Older builds left files called `emu-desktop.exe` and `emu-headless.exe` in `target/release/`. They are outdated (they do not know the newer machines); use `mimic-desktop.exe` and `mimic-headless.exe`, or start through `cargo run` as shown below, which always builds the current version first.

2.3 Starting a machine

Two equivalent ways to start the desktop emulator:

```
cargo run --release -p emu-desktop -- --machine c64 --rom-dir roms
.\target\release\mimic-desktop.exe --machine c64 --rom-dir roms
```

Everything after `--` goes to the emulator. Some first steps:

```
# C64 with a disk, loaded and started automatically
mimic-desktop --machine c64 --rom-dir roms --disk game.d64 --autostart

# Amiga 500 booting Workbench 1.3 from an ADF
mimic-desktop --machine amiga --rom-dir roms --disk workbench13.adf

# NES and SNES need no system ROM
mimic-desktop --machine nes --cart game.nes
mimic-desktop --machine snes --cart game.sfc

# Vectrex with a cartridge
mimic-desktop --machine vectrex --rom-dir roms --cart "Scramble (1982).vec"
```

`mimic-desktop --help` lists every option with the key assignments.

3 ROMs

Most machines need their original system ROMs (KERNAL, BASIC, Kickstart, operating system, BIOS). They are copyrighted and therefore **not** part of mimic: you supply them from your own machines. Put them in a folder and pass it with `--rom-dir DIR`, or set `EMU_ROM_DIR` once.

3.1 How ROMs are found

- mimic looks in the machine's subfolder first (for example `roms/c64/`), then in `roms/drives/`, then in the folder itself.
- A file is recognised by the **start of its name** (case does not matter; `.` and _ count as `-`) and by its **exact size**.
- If a ROM is missing, the error message names the role, the size and the accepted names.
- With `mimic-headless` a single ROM can also be given directly: `--rom kernal=path/kernal.bin`.

3.2 ROM table

Folder	Machine	ROMs (example names and sizes)
`c64/`	C64	`kernal` (8 KB), `basic` (8 KB), `chargen` or `characters` (4 KB)
`vic20/`	VIC-20	KERNAL, BASIC, character ROM
`pet/`	PET	BASIC, editor, KERNAL and character ROM for each model
`plus4/`	C16, Plus/4	KERNAL, BASIC; optional 3-plus-1
`c128/`	C128	KERNAL, BASIC, character ROM; optional function ROM
`atari/`	Atari 8-bit	XL OS `atarixl.rom` (16 KB), BASIC `ataribas.rom` (8 KB); for the 400/800 OS-B `atariosb.rom` (10 KB)
`cpc/`	Amstrad CPC	`os464`/`basic1-0`, `os664`/`basic664`, `os6128`/`basic1-1` (16 KB each) or 32 KB combined files; `amsdos`, optional `parados`
`amiga/`	Amiga	Kickstart images of 256 or 512 KB, e.g. `Kickstart 1.3.rom`, `kick13.rom`; one per version (1.2, 1.3, 2.0, 3.0, 3.1), chosen with `--option kickstart=`; optional A1000 bootstrap ROM (8 KB)
`zx81/`	ZX81	`zx81.rom` (8 KB)
`spectrum/`	ZX Spectrum	`Spec48.rom` (16 KB), `spec128.rom`, `specp2.rom` (32 KB each); optional `mf128.rom` (Multiface 128), `specif1.rom` (Interface 1)
`coleco/`	ColecoVision	`coleco.rom` (8 KB BIOS)
`snes/`	SNES	optional: the 64-byte sound boot ROM `spc700.rom`
`vectrex/`	Vectrex	the 8 KB system ROM (executive and Mine Storm); a 4 KB image of the executive alone also works, see the Vectrex section
`apple2/` (or `apple/`)	Apple II	files named by Apple part number: II 341-0001…0004, II+ 341-0011…0015 and 341-0020, IIe 342-0135 (CD) and 342-0134 (EF), enhanced IIe 342-0304 and 342-0303; character ROMs 341-0036, 342-0133, 342-0265; Disk II boot PROM 341-0027 and sequencer PROM 341-0028
`mac/`	Macintosh	`mac128k.rom` (64 KB), `mac512k.rom` (64 KB), `macplus.rom` (128 KB), `macse.rom` (256 KB), or each ROM's two chips (Apple part numbers 342-0220/0221, 342-0341/0342, 342-0701/0702)
`drives/`	Disk drives	`dos1541`, `dos1571`, `dos1581`

The Atari 2600 and the NES have no system ROM: the program is on the cartridge.

> **Note.** Many machines accept `--option roms=stub`: a tiny built-in program instead of the real ROMs. It is meant for tests (it shows a test screen or starts simple cartridges); real software needs the original ROMs.

4 The desktop emulator

4.1 Command-line options

Option	Meaning
`--machine ID`	the machine (see the table in section 1; default `c64`)
`--model NAME`	model variant, e.g. `ntsc`, `130xe`, `1200`
`--rom-dir DIR`	ROM folder (default: `EMU_ROM_DIR`)
`--option KEY=VALUE`	machine option, repeatable (section 5)
`--prg FILE`	a program (PRG, T64, Atari XEX) to load and run
`--disk FILE`	a disk image; repeat for further drives, extra disks form a swap list
`--tape FILE`	a tape image (TAP, CAS, CDT/TZX, ZX81 P, Spectrum TAP/TZX)
`--cart FILE`	a cartridge or console ROM
`--autostart`	load and run the first program on the disk after boot
`--crt PROFILE`	monitor look (section 7; default: the machine's own monitor)
`--scale N`	initial window size in screen rows per picture line (default 2; reduced automatically so the window fits the screen)
`--deinterlace weave|bob`	Amiga interlace: both fields woven (default) or the current field's lines doubled
`--joykeys cursor|numpad|off`	which keys play the joystick
`--joyport N`	which joystick port the keys and the gamepad play (1-based)
`--footer`	start with the status footer shown
`--no-panel`	start with the console panel hidden
`--rewind-mb N`	memory for rewinding with Ctrl+R (default 128 MB; 0 switches it off)
`--cheats DIR`	cheat database: `DIR/<machine>/<crc32>.toml` applies to the images you insert
`--cheat FILE`	a cheat file: `.toml`, a POKE list (`.pokes`), VICE monitor commands (`.mon`) or address/value codes (`.txt`)
`--input FILE`	a joystick mapping (default: the one saved with SAVE MAPPING)
`--net on|off`	the host network for the C64's WiC64 and Ultimate command interface (default on; off: they report no network)
`--uci-dir DIR`	the folder of the Ultimate command interface's DOS targets (default `./ultimate`, created on the first write)
`--debug [ADDR]`	serve the debug protocol (default 127.0.0.1:6510)
`--preview [ADDR]`	serve the live preview (default 127.0.0.1:6580)
`--vice [ADDR]`	serve the VICE binary monitor (default 127.0.0.1:6502)

4.2 Hotkeys

Key	Action
Ctrl+P	show or hide the console panel
Ctrl+Space	show or hide the status footer
Ctrl+D	put the next disk of the `--disk` list into the first drive
Ctrl+I	Amiga: switch the deinterlacing between weave and bob
Ctrl+R (held)	run backwards at twice real speed through the last frames; on release the machine runs on from there

Key	Action
F12	switch between the machine's screens (C128: 40-column VIC and 80-column VDC)
F9 / Shift+F9	datasette or cassette: PLAY / RECORD
F10 / F11	datasette or cassette: STOP / REWIND
Middle mouse button	release a captured mouse

Dropping an image file onto the window inserts it into the first drive (disks) or the cartridge slot (`.vec`, `.crt`, `.car`, `.a26`).

4.3 The console panel

Below the picture, a bar shows the buttons and switches of the machine's case: POWER and RESET, the Atari 2600 console switches, START/SELECT/OPTION of the 800XL, RESTORE of the C64 and VIC-20, 40/80 and CAPS LOCK of the C128, and so on. A button is pressed for as long as you hold the mouse button on it; clicking a switch position moves the switch.

- **PAUSE** (far left) halts the whole machine and lets it run again. It is lit while the machine stands — also when a debugger halted it — and the title bar shows where.
- **JOY KEYS** chooses the keys that play the joystick: CURSOR (cursor keys, Space fires), NUMPAD (8/2/4/6, fire on 0 or right Ctrl) or OFF (gamepad only).
- **JOY PORT** (machines with several joystick ports) moves the keyboard joystick and the gamepad to another port.
- **PAD FIRE** chooses the gamepad button that fires (SOUTH is A on an Xbox pad); **SAVE MAPPING** keeps JOY KEYS, JOY PORT and PAD FIRE for the machine, so the next start uses them (`--input FILE.toml` loads another).
- **Cheats** (from `--cheat`, the cheat database or the debugger) appear as OFF/ON switches.

4.4 Start profiles

A **start profile** stores a whole machine setup under a name: the machine and model, its options (memory, drives, slots, port devices), the ROMs, the disks and other media, the monitor, window and joystick settings and the debugger and preview servers. All profiles live in one YAML file, `profiles.yaml`, looked for in the working folder and then in the user's configuration folder (`%APPDATA%\mimic` on Windows, `~/.config/mimic` elsewhere); `--profiles FILE` names another. Its `settings` part holds what applies to every profile: the ROM folder (`roms`), where snapshots go (`snapshots`) and the folder relative disk paths count from (`media`). `docs/profiles.example.yaml` shows every key.

Command	Effect
`mimic-desktop --profile mac-se-755`	start a profile; flags given as well override it (`--profile c64 --model ntsc`)
`mimic-desktop`	no machine and no profile given: the window lists the profiles, a click starts one
`mimic-desktop --list-profiles`	list the profiles
`... --save-profile NAME`	store the setup of this command line (with a profile, its values too) as a profile; a new name is added at the end of the file, replacing an existing one rewrites the file

`mimic-headless` takes the same profile flags (without the window settings); the browser frontend reads a profile file the page loads.

4.5 The status footer

Ctrl+Space shows a status line below the picture. On the left is a **speed control**: click a step or drag the knob for 10 %, 25 %, 50 %, 100 %, 150 %, 200 %, 400 % or 800 % of real time (sound is muted away from 100 %). Next to it the **snapshot** button saves the picture as a PNG file named after the machine (`mac-001.png`, `mac-002.png` …) in the `snapshots` folder of the profile file's settings, else in the folder the emulator was started from; the footer shows the file name for a few seconds. It then shows the emulated time since power-on (m:ss.t) with the frame number, the mouse position in picture pixels, the joystick directions and the drives' activity and errors.

4.6 Window and picture

- The window opens at the largest scale up to `--scale` that fits on the screen and can be resized freely.
- The display runs in step with the monitor when its refresh rate matches the machine's (50 or 60 Hz); otherwise the emulator follows the wall clock. The console prints which mode is used.
- The window title shows the machine, the inserted disk and, while paused, where the machine stopped.

4.7 Joysticks and gamepads

Every machine with joystick ports gets a joystick played by the keyboard and by the first gamepad. The default port is the machine's game port: port 2 on the C64 and C128, port 1 on the VIC-20, C16/Plus/4, the Ataris, the CPC and the consoles, game port 1 on the Amiga (port 0 holds the mouse).

Keyboard joystick

Setting	Keys	Default on
CURSOR	cursor keys, Space fires	Atari 2600, ColecoVision, NES, SNES, Vectrex (machines without a keyboard)
NUMPAD	numeric keypad 8/2/4/6, fire on 0 or right Ctrl	the computers
OFF	gamepad only	Amstrad CPC (its keypad is the function keys)

Gamepads (Xbox 360 and others)

Gamepads are read through the operating system's game controller interface: on Windows Windows.Gaming.Input (Xbox 360, Xbox One and compatible pads), on Linux evdev, on macOS IOKit. The desktop prints the pads it finds when it starts and whenever one is plugged in or out:

```
gamepad 0 connected: Xbox 360 Controller for Windows
gamepad 1 connected: HID-compliant device
```

Gamepad 0 plays the joystick port. Buttons are named by position (South = A on an Xbox pad, East = B, West = X, North = Y):

Gamepad	Joystick and controllers
D-pad, left stick	directions (the stick is digital; on the Vectrex it is analog, see below)
South (A)	fire / button 1
East (B)	second fire button (NES B, SNES A, Vectrex button 2 …)
West (X), North (Y)	further buttons (SNES Y and X, Vectrex buttons 3 and 4)
Shoulder buttons	SNES L and R
Back / Start	Select / Start (NES, SNES)

> **Note.** Hot-plugging is fine: a pad can be connected while the emulator runs. If your pad shows up with a number other than 0, unplug other game devices or start with the pad already connected.

4.8 Mouse, paddles and light guns

When a port holds a mouse, paddles, a steering wheel or a light gun, a click into the window captures the host mouse for the machine; the middle mouse button releases it again. Examples:

- C64: `--option port1=mouse1351` (1351 mouse) or `port1=paddles`.
- Amiga: the mouse sits in game port 0 by default.
- CPC: `--option joy1=amx` (AMX mouse).
- Atari 2600: `--option port1=paddles` or `port1=driving` — the mouse turns paddle 0 or the wheel.
- NES: `--option port2=zapper` (aim with the mouse, left button fires) or `port2=arkanoid` (the mouse turns the knob).
- SNES: `--option port2=superscope` (left button = trigger, right = cursor) or `port1=mouse`.
- ColecoVision: `superaction` (Speed Roller), `driving` (wheel, left button = gas), `roller` (trackball).

4.9 Changing disks

`--disk` can be given several times. The disks go into the drives in order (Amiga: DF0:, DF1: … with `--option drives=N`); the rest form a swap list for the first drive. **Ctrl+D** inserts the next disk of that list, and an image dropped onto the window goes into the first drive as well. As when swapping by hand, the drive stays empty for about two seconds before the new disk goes in: programs that only look at the drive now and then notice the change only if they find it empty once. The title bar shows the disk in the drive.

```
# a demo on two disks with one drive: press Ctrl+D when it asks for disk B
mimic-desktop --machine amiga --rom-dir roms --disk demo-a.adf --disk demo-b.adf
```

5 The machines

Keyboards are mapped by **position** on a PC keyboard (US layout), not by the printed labels: the key where the original machine has a key plays that key. Options are set with `--option KEY=VALUE` and can be repeated.

5.1 Commodore 64

Option	Values
`sid`	`6581` or `8580`
`cia`	`6526` or `6526a`
`drive8`	`1541` (true drive emulation), `1571`, `1581`, `virtual` (fast, KERNAL load/save only), `none`
`port1`, `port2`	`joystick`, `paddles`, `mouse1351`, `none`
`autostart`	`on`: load and run the disk's first program
`uci`	`on`: the Ultimate 64 command interface at $DF1C-$DF1F (see below)
`userport`	`wic64`: a WiC64 WLAN adapter in the user port; `none` (default)
`net`	`off`: no network for the UCI and the WiC64, even if the frontend has one

Keys: Esc = RUN/STOP, Tab = CTRL, left Ctrl = C=, Insert = £, Delete = ↑, Page Up = RESTORE, Home = CLR/HOME, F2/F4/F6/F8 = shifted F1/F3/F5/F7. The joystick is in port 2.

A **folder** can be used as a disk with the virtual drive: `--option drive8=virtual --disk ./myprograms`.

Network: Ultimate command interface and WiC64

Ultimate command interface (`--option uci=on`): the four registers of the Ultimate 64 / 1541 Ultimate II+ command interface at $DF1C-$DF1F. The DOS targets 1 and 2 work on a folder of your computer (`--uci-dir`, default `./ultimate`): open, read, write, seek, file info, delete, rename, copy, change, create and list directories, the time. Paths can never leave the folder. The network target 3 opens TCP and UDP sockets (and a TCP listener on 127.0.0.1); the control target 4 answers IDENTIFY. The Ultimate's own control commands (mounting, reset, REU, palette) are not offered. A cartridge can stay plugged in; the interface takes only its four registers.

WiC64 (`--option userport=wic64`): the WLAN adapter on the user port with firmware 2.x commands in the legacy and the protocol-2 format: HTTP GET (plain and URL-encoded), HTTP POST, TCP open/read/write/close, IP, MAC, SSID, signal strength, time and time zone, server, version, status message, time-outs and echo. `https:` addresses are reported as unsupported.

Both use your computer's network by default; `--net off` switches it off and the devices report "no network". The clock is UTC unless the environment variable `EMU_UTC_OFFSET` (e.g. `+01:00`) gives the offset. The web version has no network (browsers allow no sockets); there the command interface sees the uploaded files as its folder. A debugger recording (`record_start`/`record_stop`) also records every answer from the network and the folder, so `replay_input` repeats a run exactly without the network.

5.2 Commodore VIC-20, C16/Plus/4, PET and C128

Machine	Options
VIC-20	`ram=none|3k|8k|16k|24k|all`, `port1=joystick|none`, `drive8=…`, `autostart=on`
C16/Plus/4	`ram=16k|64k`, `drive8=…`, `port1=joystick|none`, `autostart=on`
PET	`ram=8k|16k|32k`, `keyboard=graphics|business`, `hz=50|60`, `drive8=virtual|none`
C128	`display=40|80`, `vdc_ram=16k|64k`, `sid=…`, `drive8=1571|1541|1581|virtual|none`, `port2=…`, `autostart=on`

The C128 has two screens: F12 switches between the 40-column VIC and the 80-column VDC; the window starts on the one the 40/80 key selects.

5.3 Atari 400, 800, 800XL, 65XE and 130XE

Option	Values
`port1`, `port2`	`joystick`, `paddles`, `none` (400/800 also `port3`, `port4`)
`basic`	`on`/`off` — XL/XE: hold OPTION at boot; 400/800: BASIC cartridge in the left slot
`ram`	400/800 only: `8k`, `16k`, `32k`, `48k`
`drives`	`0` to `4` disk drives
`drive`	`1050`, `810` or `xf551`

Keys: < > on - =, - = on [ ], + *on* ' \, *right Alt = Atari key, cursor keys = CONTROL* + - = + , F2 START, F3 SELECT, F4 OPTION, F5 RESET, F6 HELP, F7/Pause BREAK.

5.4 Atari 2600

Option	Values
`port1`, `port2`	`joystick`, `paddles`, `keypad`, `driving`, `none`
`tv`	`color` or `bw`
`left`, `right`	difficulty switches `b` or `a`
`bank`	force a bank-switching scheme: `2k`, `4k`, `f8`, `f6`, `f4`, `f8sc`, `f6sc`, `f4sc`, `fa`, `fe`, `e0`, `e7`, `3f`, `dpc` (Pitfall II; normally detected)

Console: F1 SELECT, F2 RESET (held), F3 colour/black-and-white, F4/F5 left/right difficulty. Keypads: left 1 2 3 / Q W E / A S D / Z X C, right 8 9 0 / I O P / K L ; / , . /.

5.5 Amstrad CPC 464, 664 and 6128

Option	Values
`crtc`	`0`, `1` or `2` (default: 0 on the 464, else 1)
`dos`	`amsdos`, `parados` or `none`
`joy1`	`joystick`, `amx` (AMX mouse) or `none`
`joy2`	`joystick` (a second stick daisy-chained) or `none`

Disks (DSK/EDSK) go into drive A on the 664/6128: type `CAT` or `RUN"DISC`. The 464 has a cassette only (CDT/TZX); on the 664/6128 type `\|TAPE` first to switch to tape. Keys: - ^ on - =, @ [on [ ], : ; on ; ',] on the backslash key, Backspace = DEL, Delete = CLR, left Alt = COPY, the numeric keypad and F1–F10 = the function keys f0–f9 and f.

5.6 Commodore Amiga

Option	Values
`kickstart`	`1.2`, `1.3`, `2.0`, `3.0`, `3.1` (default 1.3; A500+ 2.0; A1200 3.1)
`chip`	chip RAM `256k`, `512k`, `1m`, `2m`
`slow`	trapdoor RAM `512k` or `0`
`fast`	Zorro II fast RAM `0` … `8m` (found by Kickstart through autoconfig)
`chipset`	`ocs`, `ecs`, `aga`; or `agnus=ocs|ecs`, `denise=ocs|ecs` separately
`drives`	`1` to `4` (DF0: to DF3:)

Option	Values
`drivetype`	`dd` or `hd` (1.76 MB disks)
`port0`, `port1`	`mouse`, `joystick`, `none` (defaults: mouse in 0, joystick in 1)
`serial`	`none` or `loopback`
`wcs`	A1000: `rom` or `disk` (load Kickstart from disk with the bootstrap ROM)

Without a disk, Kickstart shows its insert-disk screen; an ADF in DF0: boots. Workbench 1.3 takes about three minutes of emulated time to reach its screen from disk. Keys: left/right Windows key = left/right Amiga key, both Ctrl = CTRL, Insert = HELP, Delete = DEL; **Ctrl + both Windows keys** resets like the real keyboard. Sound is stereo with the A500's filters.

5.7 Sinclair ZX81

Options: `ram=16k|1k|2k` (default 16k), `traps=on|off`. Programs are `.p` files given with `--tape`: `LOAD ""` loads the program from the tape slot and `SAVE "NAME"` writes it, through traps in the ROM.

5.8 Sinclair ZX Spectrum

Models `48k`, `128k` and `plus2`. Options: `traps=on|off` (default on: `LOAD ""` copies standard tape blocks at once; off plays the tape in real time), `autostart=on` (with a tape: types `LOAD ""`, or picks the 128K menu's Tape Loader), `joystick=kempston|sinclair1|sinclair2|cursor|none`, `mouse=kempston`, `issue=3|2`, `multiface=on`, `if1=on` with `microdrives=1..8`, `roms=stub`.

Tapes are TAP or TZX files (`--tape`); `LOAD ""` presses PLAY, and the tape stops at TZX "stop" blocks. Snapshots (SNA, Z80) go in with `--prg` and take over at once. Keys: Shift = CAPS SHIFT, Ctrl or AltGr = SYMBOL SHIFT, Backspace = DELETE, the cursor keys = CAPS SHIFT + 5–8, Escape = BREAK, `.` and `,` as on the keys. The Multiface 128's red button is the panel's MULTIFACE; the Interface 1's Microdrives take MDR cartridges (`--disk`, `FORMAT "m";1;"name"`, `SAVE *"m";1;"file"`, `CAT 1`, `LOAD *"m";1;"file"`). The RS-232 port has no host device yet.

5.9 ColecoVision

Options: `port1`, `port2` = `controller` (default: joystick, two fire buttons, keypad), `superaction`, `driving`, `roller`, `none`; `roms=stub`. Keys: controller 1's keypad on the digit row (1–9, 0, - = *, = = #, [ ] = the Super Action Controller's purple and blue buttons, X = right fire), controller 2's keypad on the numeric keypad; joystick 1 on the cursor keys, Space = left fire.

5.10 Nintendo NES / Famicom

Options: `port1`, `port2` = `controller`, `zapper`, `arkanoid`, `none`; `fourscore=on` (ports 3 and 4). Keys: the D-pad on the cursor keys, Space and X = A, Z = B, right Shift = Select, Enter = Start. Cartridges are iNES or NES 2.0 files with mappers 0–4 (NROM, MMC1, UxROM, CNROM, MMC3).

5.11 Super Nintendo / Super Famicom

Options: `port1`, `port2` = `pad`, `mouse`, `superscope` (port 2 only), `none`; `multitap=on` (Multiplayer 5 in port 2); `roms=stub`. Keys: the D-pad on the cursor keys, Z = B (also Space), X = A, A = Y, S = X, Q = L, W = R, right Shift = Select, Enter = Start. Cartridges are `.sfc`/`.smc` (LoROM, HiROM, ExHiROM, with or without copier header); games with coprocessor chips (DSP, Super FX, SA-1) are not supported yet.

5.12 Vectrex

The Vectrex draws with a beam steered by analog hardware, not with pixels. mimic follows the beam and shows a rendered picture of it in the window (portrait, 800 × 1000 pixels).

Option	Values
`port1`	`controller` (default) or `none`
`port2`	`controller` (default), `lightpen`, `imager` (3D Imager) or `none`
`lines`	`fine` (default: thin lines, bright points where the beam rests) or `soft` (even, brighter lines with a soft edge)
`eye`	with the 3D Imager: `left` (default), `right` or `both`
`roms`	`stub`: a test picture without the system ROM

Controls: the analog stick on the gamepad's left stick (analog) or the cursor keys and D-pad (full deflection); buttons 1–4 on Z X C V or A B X Y of the gamepad; Space = button 1.

- **System ROM.** The real ROM is 8 KB: the executive and the built-in game Mine Storm. A 4 KB image of the executive alone starts cartridges, but without a cartridge there is no Mine Storm, and cartridges that call Mine Storm's routines (such as 3-D Mine Storm) hang.
- **Start-up.** After power-on the executive shows its title for about ten seconds before a cartridge starts.
- **3D Imager.** The spinning colour wheel is emulated (motor, index pulse, colour filters); the picture shows the eye chosen with `eye`.

```
mimic-desktop --machine vectrex --rom-dir roms --cart "Scramble (1982).vec"
mimic-desktop --machine vectrex --rom-dir roms --option lines=soft --cart game.vec
mimic-desktop --machine vectrex --rom-dir roms --option port2=imager --cart 3d-game.vec
```

5.13 Apple II

The Apple II family as four models: the original II (Integer BASIC), the II+ (Applesoft), the IIe and the enhanced IIe (65C02, MouseText). Every model has a language card, a Disk II card with two drives in slot 6 and a Mockingboard sound card in slot 4. Colour comes from the monitor: the Apple sends a black-and-white dot pattern and a colour TV decodes colours from it, so the default monitor is `tv`; text screens show in black and white (the Apple turns the colour burst off), and `--crt raw` shows the dot pattern itself.

Option	Values
`ram`	II/II+: `4k` … `48k` (default `48k`)
`lc`	II/II+: language card `on` (default) or `off`
`aux`	IIe: auxiliary memory `64k` (default, Extended 80-Column Card), `1k` or `none`
`slot4`	`mockingboard` (default) or `none`
`slot6`	`disk2` (default) or `none` (without it the machine starts BASIC instead of waiting for a disk)
`port`	game I/O: `joystick` (default), `paddles` or `none`
`roms`	`stub`: a test ROM without the Apple ROMs

Keyboard: the PC keys type what their legends show (letters in upper case on the II and II+); Backspace and ← are the Apple's ← (backspace), the Alt keys are Open Apple and Solid Apple, **F5 is RESET** (on the IIe with Ctrl, as on the real machine). The joystick is analog: the gamepad's left stick, or the numeric keypad at full deflection.

- **Disks.** DSK/DO (DOS 3.3 order), PO (ProDOS order), NIB and WOZ 1/2 images go into slot 6, drive 1 (`--disk`; a second `--disk` fills drive 2). With the boot PROM the machine boots the disk at power-on; without a disk it waits with the drive running, like the real one.

- **Cassette.** WAV recordings (`--tape`). The tape moves while a program listens to it, so type `LOAD` (or the monitor's `R`) at any time. To save, insert an empty file, press Shift+F9 (RECORD), `SAVE`, then F10 (STOP): the recording is written as a WAV file.
- **Sequencer PROM.** With the P6 PROM (341-0028) the Disk II reads and writes the way the real card does; without it a model of the card stands in.

```
mimic-desktop --machine apple2 --rom-dir roms --disk "Zork I.dsk"
mimic-desktop --machine apple2 --model iiplus --rom-dir roms --disk game.woz
mimic-desktop --machine apple2 --model ii --option slot6=none --rom-dir roms
```

5.14 Macintosh

The compact Macintosh as four models: the 128K and 512K (64 KB ROM, 400K single-sided drives, the M0110 keyboard without keypad), the Plus (128 KB ROM, 800K drives, SCSI, the M0110A keyboard with keypad and arrows) and the SE (256 KB ROM, ADB keyboard and mouse, two internal drives, SCSI). Without a disk the Mac shows the disk icon with a question mark, as the real one does. The screen is the built-in black-and-white monitor (profile `mac`).

Option	Values
`ram`	Plus/SE: `512k`, `1m` (default), `2m` or `4m`
`serial`	`loopback`: both serial ports wired back to themselves
`roms`	`stub`: a test ROM without the Mac ROMs

Keyboard: the PC keys sit where the Mac's are on a US layout; **Alt is Command** (next to the space bar, as on the Mac), the Windows keys are Option, NumLock is the keypad's Clear. The mouse is captured with a click into the window. The panel has POWER, RESET and the programmer's INTERRUPT switch.

- **Floppies.** 400K and 800K images, raw (`.dsk`, `.img`) or DiskCopy 4.2 (`.image`), with tag bytes; written back in their own format. The first `--disk` goes into the internal drive, the next into the external one (the SE: the second internal drive first).
- **Hard disks.** A hard disk image (whole 512-byte blocks, with an Apple partition map and driver, as Apple's HD SC Setup makes it) given with `--disk` goes to the first SCSI ID that is free: a floppy drive refuses it and passes it on (Plus and SE).
- **System software.** System 6.0.8 suits the Plus (1 MB or more) and the SE; the 128K needs System 1.x/2.x on 400K MFS disks, the 512K up to 3.x.

```
mimic-desktop --machine mac --rom-dir roms --disk "System Tools.image"
mimic-desktop --machine mac --model se --rom-dir roms --disk system.dsk --disk hd20.img
mimic-desktop --machine mac --model 128k --rom-dir roms --disk mfs-system.dsk
```

6 Disks, tapes, cartridges and programs

Kind	Formats	Option
Disk	D64, D71, D81, G64, D80/D82 (PET), ATR (Atari), DSK/EDSK (CPC), ADF (Amiga DD and HD), extended ADF, DMS (read only), DSK/DO/PO/NIB/WOZ (Apple II), 400K/800K DSK and DiskCopy 4.2, SCSI hard disk images (Macintosh); a folder as a virtual disk	`--disk`
Tape	TAP (Commodore), CAS (Atari), CDT/TZX (CPC), P/81 (ZX81), TAP/TZX (ZX Spectrum), WAV (Apple II); a PRG is recorded in the machine's tape format	`--tape`
Cartridge	CRT (C64/C128), raw VIC-20 and C16/Plus/4 dumps, CAR or 8/16 KB ROM (Atari), .a26/.bin (Atari 2600), .col/.rom (ColecoVision, MegaCart), .nes (NES), .sfc/.smc (SNES), .vec/.bin (Vectrex, up to 64 KB)	`--cart`
Program	PRG, T64, XEX	`--prg`

Images are opened through a **copy-on-write overlay**: the file on your disk stays unchanged while the emulated machine writes to its copy. Changes are written back only on request (`flush_media` in the debugger).

6.1 Commodore drives

Each Commodore drive slot has two modes: **true drive emulation** (the drive's own CPU and the disk's GCR data, cycle-exact — needed for fast loaders and copy protection) and the **virtual drive** (serves files on the protocol level — fast, but only for normal LOAD and SAVE).

7 Monitor profiles

Machines always produce the raw picture; monitor and signal effects (composite colour bleeding, PAL colour mixing, NTSC artefact colours, scanlines, phosphor) are added afterwards. `--crt PROFILE` chooses the look; without it each machine uses the monitor it shipped with.

Profile	Use
`raw`	sharp pixels, no effects (default for the NES, ColecoVision and Vectrex)
`1702`, `1084s`	Commodore monitors over S-Video (C64, VIC-20, C16/Plus/4, C128)
`1084s-rgb`	the 1084S on its RGB input (Amiga, SNES, C128 VDC)
`tv`, `pal-tv`, `ntsc-tv`	a television over composite (Atari, ZX81, ZX Spectrum, Apple II)
`pet-white`, `pet-green`	the PET's built-in monitors
`ctm644`	Amstrad colour monitor (CPC)
`gt65`	Amstrad green monitor
`mac`	the compact Macintosh's built-in black-and-white monitor
`lite`	aspect correction and light scanlines only, for slow computers and the web version

Besides the signal (composite, S-Video, RGB, monochrome) a profile sets scanlines, the phosphor mask (`shadow`, `aperture` or `slot` with strength and pitch), bloom (bright areas glow), the curvature of the glass, a vignette and the persistence of the phosphor (a trail behind moving objects, as on the green PET and GT65 monitors). The same picture comes out of the graphics card in the desktop program and of the processor in snapshots, the preview and the web version.

Own profiles are TOML files that start from a preset and change single values (`base = "1084s"` …), loaded with `--crt myprofile.toml`. The desktop program re-reads such a file when it changes, so a profile can be tuned while the machine runs. The live preview has a **monitor editor** at `http://127.0.0.1:6580/crt`: sliders for every value, the machine's picture through the profile, and a button that saves the TOML file.

8 The command-line program

`mimic-headless` runs a machine without a window or sound — for screenshots, scripts, automated tests and as a debug server.

```
# run 300 frames and save the last one as PNG
mimic-headless --machine c64 --rom-dir roms --prg demo.prg --frames 300 --snapshot out.png

# C128 in 80 columns, picture of the VDC
mimic-headless --machine c128 --option display=80 --rom-dir roms --frames 600 --screen vdc --snapshot c128.png

# through a monitor profile, sound to WAV, savestate afterwards
mimic-headless --machine cpc --rom-dir roms --frames 150 --crt ctm644 --snapshot cpc.png --wav cpc.wav --save-state
```

Option	Meaning
`--frames N`	frames to run (default 1)
`--snapshot FILE.png`	save the last frame
`--snapshot-every N`	with `--snapshot`: also save every N-th frame
`--screen NAME`	which screen (C128: `vic` or `vdc`)
`--crt PROFILE[:N]`	the snapshot through a monitor profile, N rows per line
`--wav FILE.wav`	the sound of the run as 16-bit PCM
`--load-state F`, `--save-state F`	restore a savestate before, write one after the run
`--rom ROLE=FILE`	a single ROM image
`--cheats DIR`	cheat database: `DIR/<machine>/<crc32>.toml` applies to the images given
`--cheat FILE`	a cheat file: `.toml`, a POKE list (`.pokes`), VICE monitor commands (`.mon`) or address/value codes (`.txt`)
`--debug [ADDR]`	serve the debugger instead of running frames
`--preview [ADDR]`	serve the live preview (implies `--debug` and `--run`)
`--vice [ADDR]`	also serve the VICE binary monitor
`--run`	with `--debug`: start running instead of paused
`--no-realtime`	with `--debug`: run as fast as possible
`--list`	list the machines

The other options (`--machine`, `--model`, `--option`, `--disk`, `--tape`, `--cart`, `--prg`, `--autostart`, `--rom-dir`) work as in the desktop program.

9 Live preview in the browser

With `--preview` the emulator serves a viewer page: open `http://127.0.0.1:6580/` in a browser to watch the running machine. The viewer forwards keyboard, gamepad and mouse input, so the machine can be played from the browser, and shows the console panel.

```
mimic-headless --machine c64 --rom-dir roms --preview
mimic-desktop  --machine amiga --rom-dir roms --preview      # window and browser
```

Address	Content
`/`	the viewer page
`/snapshot.png`	the current frame as PNG
`/preview` (WebSocket)	a picture stream as PNG or QOI, also through a monitor profile
`/stream` (WebSocket)	a raw stream: colour indices, changed lines only, optional sound
`/controls`	the console's buttons and switches
`/gfx`, `/gfx/VIEW.png`	the graphics the video chip works from: sprites, character sets, bitmaps, screen memory, tiles, palettes
`/overlays`	the debug overlays the machine offers
`/crt`	the monitor profile editor

Parameters: `filter=raw|crt`, `profile=NAME`, `scale=1..4`, for the streams `fps=1..50`, `format=png|qoi`, `delta=0|1`, `audio=0|1`. The server listens on 127.0.0.1 only.

Overlays (`overlay=beam,grid,…`, switches in the viewer's header) draw debugging information over the raw picture: the beam position (also in a picture stopped half way), a raster grid with line numbers, and what the video chip did where — the C64's bad lines and where each sprite really appeared, the Atari display list's mode lines, the Amiga's DMA slot map and Copper writes, CPC CRTC changes, NES and SNES sprites, interrupts and HDMA. The **Graphics data** panel under the picture shows the sprites, character sets, bitmaps and other graphics in memory; its options browse other addresses (`index=2 addr=$2000`).

10 Debugging

Every machine can be debugged the same way. The debugger knows no particular machine: it works through registers, memory spaces, chips and devices that each machine describes, so second processors (the 1541's 6502, the C128's Z80, the Amiga's Copper and Blitter, the SNES sound CPU) are debug targets of their own. Looking at memory never changes it — reading an interrupt register in the debugger does not acknowledge the interrupt.

10.1 Starting the debug server

```
mimic-headless --machine c64 --rom-dir roms --debug          # paused, port 6510
mimic-desktop  --machine c64 --rom-dir roms --debug          # with window and sound
mimic-headless --machine c64 --rom-dir roms --vice --preview --run
```

10.2 The Python client polydbg

`tools/polydbg.py` needs only Python's standard library:

```
python tools/polydbg.py info                # devices and memory spaces
python tools/polydbg.py regs
python tools/polydbg.py disasm
python tools/polydbg.py break '$0810'       # breakpoint
python tools/polydbg.py resume --wait       # run until it stops
python tools/polydbg.py peek '$d000' 64
python tools/polydbg.py chips vic2          # chip registers
python tools/polydbg.py freeze '$0400' 9    # freeze a memory value (cheat)
python tools/polydbg.py search start        # cheat search
python tools/polydbg.py save state.bin      # savestate
python tools/polydbg.py gfx sprites index=2 --scale 4 -o s.png   # graphics in memory
python tools/polydbg.py picture --overlays beam,badlines -o now.png
python tools/polydbg.py rewind --on 1       # keep a state every frame
python tools/polydbg.py rewind 50           # 50 frames back
python tools/polydbg.py back 3              # 3 instructions back
python tools/polydbg.py shell               # interactive prompt
```

`--host`/`--port` choose the server, `--json` prints raw answers; `python tools/polydbg.py --help` lists all commands.

10.3 What the debugger can do

- **Execution:** pause, resume, step an instruction, a cycle, a raster line or a frame, step over and out, run to an address.
- **Breakpoints:** on execution, memory reads and writes, raster positions, IRQ and NMI; with conditions such as `A == $FF && [$D012] > 100` and hit counts.
- **Inspection:** registers, memory, disassembly, chip registers, beam position; symbols from VICE label files and KickAssembler.
- **State:** savestates, a raster trace of instructions and memory accesses.
- **Backwards:** rewind by frames, and step back instruction by instruction (the emulator replays from the last saved moment, exactly).
- **Pictures:** the current picture, also half drawn, with the beam marked and overlays; the graphics data the video chip works from.
- **Cheats:** freeze values in memory, narrowing search for unknown addresses (lives, energy), cheat files with code patches, imports of POKE lists, VICE monitor commands and address/value codes, and a cheat database that applies by the image's CRC-32.
- **Devices and media:** plug devices into ports, insert and eject media, inject and record input.

10.4 VICE binary monitor (VS64 and others)

With `--vice` mimic also speaks the binary monitor protocol of the VICE emulator. Tools such as VS64 (Visual Studio Code), C64 Studio or IceBro Lite can then debug any Commodore 8-bit machine in mimic. In VS64's `launch.json`:

```
{ "type": "vice", "request": "attach", "name": "mimic",
  "hostname": "127.0.0.1", "port": 6502 }
```

> **Note.** Use `127.0.0.1`, not `localhost`: the servers listen on IPv4 only.

11 The web version

Every machine also runs in the browser, compiled to WebAssembly. Nothing is bundled: the ROMs and images come from your own files, which the page can keep in the browser for the next visit.

```
bash tools/build-web.sh                                  # builds target/web
python -m http.server -b 127.0.0.1 -d target/web 8080    # then open http://127.0.0.1:8080/
```

- **Machine:** choose the machine, model and options, then Start. "built-in test ROMs" runs without ROM files.
- **Files:** add single files or a whole ROM folder (the same names and folders as `--rom-dir`). Files dropped onto the picture go into the matching drive.
- **Drives and slots:** insert and eject disks, tapes and cartridges.
- **Cheats:** switch freezes on and off, import POKE lists and codes, search memory and freeze what you found; cheat files `cheats/<machine>/<crc32>.toml` in an added folder apply automatically.
- **State:** download and load savestates; start a profile from a `profiles.yaml`.
- Click the picture to type into the machine; gamepads play the joystick; a mouse in a port captures the pointer (Esc releases it).
- A link such as `?machine=c64&options=roms=stub&start=1` starts a machine at once.

12 Troubleshooting

Problem	Solution
`unknown machine 'vectrex'` (or another new machine)	You started an outdated `emu-desktop.exe`. Use `mimic-desktop.exe` or `cargo run --release -p emu-desktop --` ….
`missing ROM '…'` or `no … ROM`	Check the folder given with `--rom-dir` and the file's exact size; the message names the accepted file names.
The emulator runs slowly or stutters	Use the release build. `EMU_DESKTOP_DEBUG=1` prints frame and sound statistics every 5 seconds. Check the speed control in the footer (Ctrl+Space).
The window is too large or too small	The window fits itself to the screen; resize it, or set `--scale 1`/`--scale 3`.
The gamepad does nothing	Look at the console output: the pad must appear as `gamepad 0`. Check JOY PORT in the panel and that the machine has a joystick in that port.
The keyboard types instead of steering	Set JOY KEYS in the panel (or `--joykeys`) to CURSOR or NUMPAD.
A disk program waits for the next disk	Give all disks with `--disk` and press Ctrl+D, or drop the image onto the window.
Vectrex: black screen after the cartridge's title	The cartridge may need the full 8 KB system ROM (it calls Mine Storm's routines), or a 3D Imager (`--option port2=imager`).
No sound on Linux	Install `libasound2-dev` before building.

12.1 Getting help

- `mimic-desktop --help` and `mimic-headless --help` list every option and key.
- `docs/usage.md` is the detailed reference (German) for building, ROMs, media, the debug protocol and the test suites.
- `README.md` lists all CPUs, machines, chips and devices with their state.